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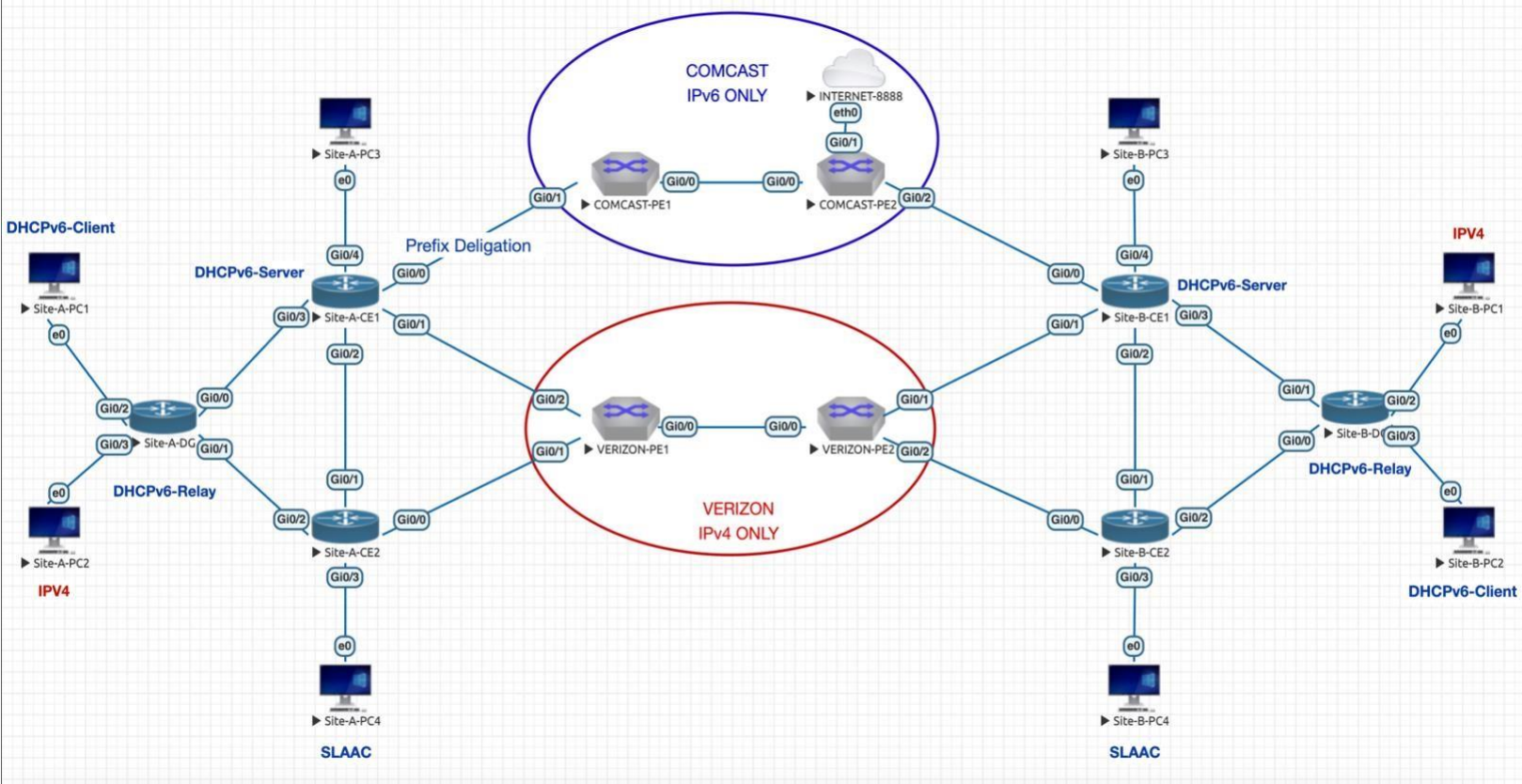
Introduction to Enterprise Networks

Spring 2025

Lab 6

IPv6 - IPv6 Tunneling Mechanisms

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IPv6 Transition Lab: Global Tech's Dual-Stack Migration

1. Introduction

As IPv6 adoption becomes essential for modern networks, **Global Tech** has begun transitioning its corporate sites—**Site-A** and **Site-B**—from **IPv4 to IPv6**. However, due to budget constraints, the migration will be **incremental**, meaning that some infrastructure at both sites will still operate on IPv4 during the transition period.

To maintain **full reachability** between IPv4 and IPv6 nodes across both sites and provide **internet access**, a combination of **dual stack, tunneling, and dynamic IPv6 addressing techniques** will be implemented.

2. Network Overview

- **Sites:**
 - **Site-A:** Uses **OSPFv3** for IPv6 routing.
 - **Site-B:** Uses **RIPng** for IPv6 routing.
- **ISPs:**
 - **Comcast (Primary ISP)** – IPv6-only backbone.
 - **Verizon (Backup ISP)** – IPv4-only backbone.
- **IPv6 Addressing:**
 - **Prefix Delegation (PD), SLAAC, and DHCPv6** will be used to automate IPv6 assignment.
- **Internet Connectivity:**
 - All clients in **Site-A** and **Site-B** should have IPv6 **internet access** via **Comcast**, which will be simulated using a **loopback interface on Comcast's PE-2 router**.
- **IPv4-IPv6 Coexistence:**
 - Various **tunneling mechanisms** will ensure communication between IPv4-only and IPv6only nodes:
 - GRE (Generic Routing Encapsulation)
 - 6to4 (Automatic IPv6 over IPv4)
 - 4over6 (IPv4 over IPv6 transport)

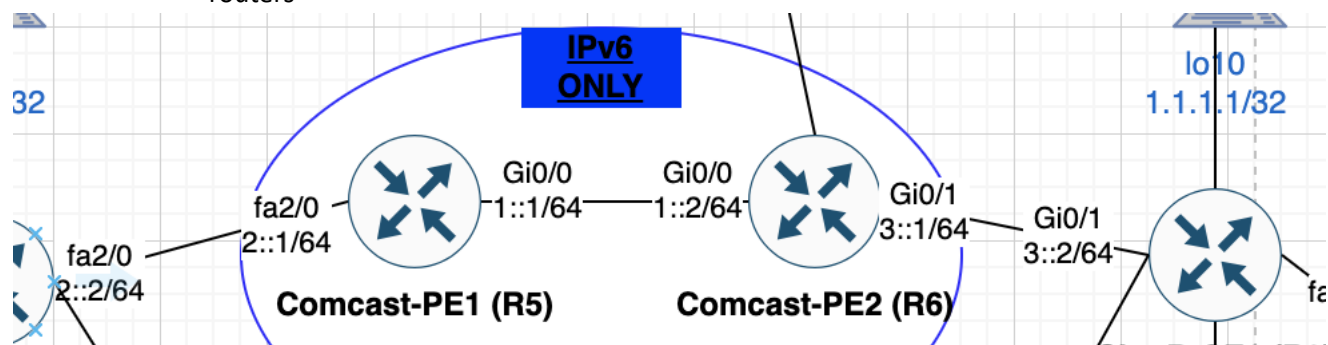
3. Lab goals

- Ensure full reachability between IPv4 and IPv6 nodes in both sites.
- Implement dual-stack routing and IPv6 address assignment using Prefix Delegation, SLAAC, and DHCPv6.
- Use OSPFv3 in Site-A and RIPng in Site-B for intra-site IPv6 routing.
- Establish IPv6 internet access for all clients via Comcast.
- Maintain IPv4 connectivity across the IPv6-only ISP (Comcast) using tunneling techniques.

4. Objectives

Objective 1: Comcast

- Assign manual IPv6 addresses to comcast PE1 and PE2. IPv4 is not supported on these routers



- Configure Comcast-PE1 as an IPv6 Prefix Delegator. This router gives a /56 prefix to SiteA for internal IPv6 assignment.

```

ipv6 dhcp pool DELEGATE_SITEA
prefix-delegation 2001:DB8:2::/56 56
interface FastEthernet2/0
no ip address
duplex auto
speed auto
ipv6 address 2::1/64
ipv6 enable
ipv6 dhcp server DELEGATE_SITEA
!

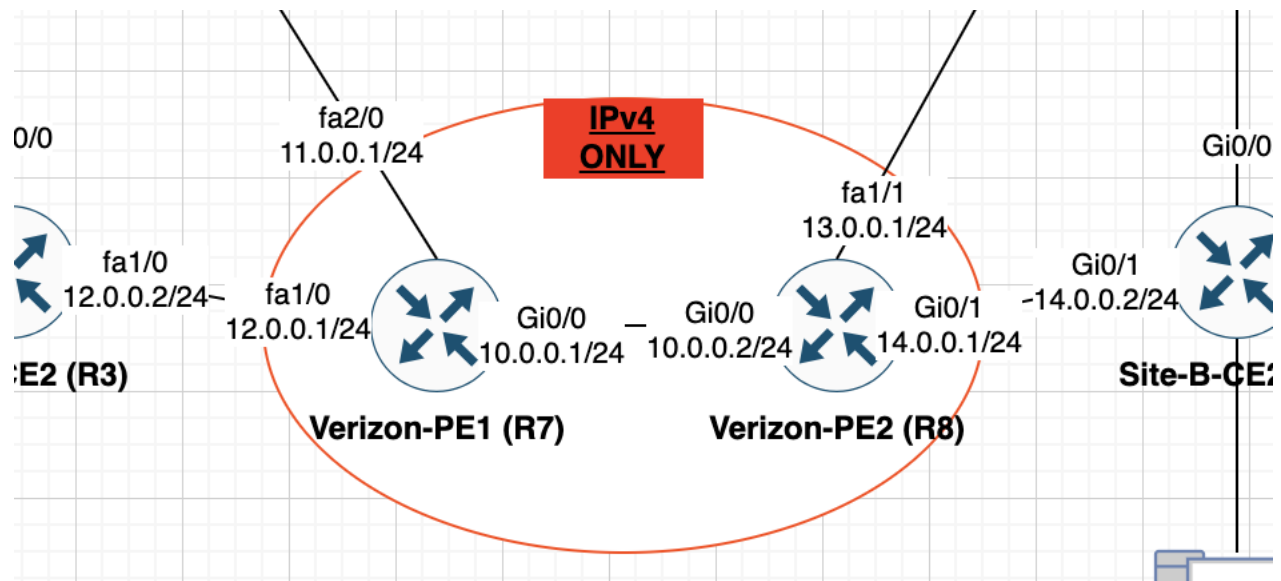
```

- Internet is an IPv6 loopback address in Comcast-PE2 with the address 2001:db8::ffff:0808:0808/8

```
interface Loopback0
no ip address
ipv6 address 2001:DB8::FFFF:808:808/8
ipv6 enable
!
```

Objective 2: Verizon

- Assign manual **IPv4** addresses to Verizon PE1 and PE2. IPv6 is not supported on these routers



- Ports connected to the Verizon PE1 and PE2 speak IPv4 only.

Objective 3: Site-A

- Comcast-PE1** is the Prefix Delegator for Site-A. Configure this router to assign a /56 prefix to Site-A-CE1 which is the prefix receiver.

```
Comcast-PE1:
ipv6 dhcp pool CUSTOMERS
prefix-delegation pool GLOBAL_POOL
domain-name NETWORKLESSONS.LOCAL
!
interface FastEthernet2/0
ipv6 address 2001:DB8:0:1::1/64
ipv6 enable
ipv6 dhcp server CUSTOMERS
!
ipv6 local pool GLOBAL_POOL 2001:DB8:1100::/40 48
```

- **Site-A-CE1** should automatically configure itself to distribute **IPv6 addresses to all internal clients** using the **/56 delegated prefix**.

```
Site-A-CE1:
interface FastEthernet2/0
ipv6 address autoconfig default
ipv6 enable
ipv6 dhcp client pd ISP_PREFIX
!
```

- In IP assignments, consider the following points:
 - Site-A-DG is a DHCPv6-Relay for Site-A, therefore, Site-A-PC1 should get its IP address from the relay.
 - Site-A-PC2 IPv4 address should be entered manually
 - Site-A-PC3 uses SLAAC method for IP assignment and receives router advertisements from Site-A-CE1.
 - Site-A-PC4 IPv6 address should be entered manually
- Run OSPFv3 on all links inside Site-A

Site-A-CE1 OSPFv3 config:

```
ipv6 router ospf 1
router-id 1.1.1.1
default-information originate always
interface GigabitEthernet0/0
ipv6 ospf 1 area 0
```

Objective 4: Site-B

- Unlike Site-A, there is no prefix delegation in this site. However, you need to configure DHCPv6 server and IP address assignment manually using another /56 prefix.
- Assign IPv6 addresses manually for the interfaces between Site-B-CE1 and Comcast-PE1
- Setup DHCPv6 Server on Site-B-CE1 and use its manual DHCP router advertisement for assigning IPv6 addresses to the clients. Consider the following points:
 - Site-B-PC1 IPv4 address should be entered manually
 - Site-B-PC2 gets its IPv6 address from Site-B-DG which is the DHCPv6-Relay
 - Site-B-PC3 uses SLAAC method for IPv6 assignment.
 - Site-B-PC4 IPv6 address should be entered manually
- Enable RIPng on all links inside Site-B

RIPng config for Site-B-CE1:

```
ipv6 router rip SITEA
redistribute static
interface FastEthernet2/0
ipv6 rip SITEA enable
```


Objective 5: Tunneling mechanisms

- Verify that all IPv6 PCs on both sites can ping each other via the Comcast ISP
 - Paste Screenshots of the traceroutes

Traceroute of Site A PC1 to Site B PC4:

```

Site-B-CE2#traceroute ipv6
Target IPv6 address: 13::1
Source address: 16::1
Insert source routing header? [no]:
Numeric display? [no]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Priority [0]:
Port Number [0]:
Type escape sequence to abort.
Tracing the route to 13::1

 1 9::1 0 msec 0 msec 0 msec
 2 3::1 0 msec 0 msec 0 msec
 3 1::1 0 msec 0 msec 0 msec
 4 2::2 0 msec 0 msec 0 msec
 5 4::2 0 msec 0 msec 0 msec
Site-B-CE2#

```

- In normal conditions when Comcast is available, the IPv4 traffic should be routed through Comcast via a GRE Support over IPv6 Transport tunneling mechanism. Make a GRE tunnel for IPv4 over IPv6 transport from Site-A-DG to Site-B-DG. Do a ping and traceroute from Site-A-PC2 to Site-B-PC1

Paste screenshots of the traceroute showing only two hops + commands used for building the tunnel

```

Site-A-DG#traceroute 200.0.0.1
Type escape sequence to abort.
Tracing the route to 200.0.0.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.10.10.2 0 msec 0 msec *
Site-A-DG#

```

I did it in 1 hop!

Site-A-DG config:

```
interface Tunnel0
ip address 10.10.10.1 255.255.255.0
tunnel source 13::1
tunnel mode gre ipv6
tunnel destination 15::1
!
router ospf 50
network 10.10.10.0 0.0.0.255 area 0
network 100.0.0.0 0.0.0.255 area 0
!
```

Site-B-DG config:

```
interface Tunnel0
ip address 10.10.10.2 255.255.255.0
tunnel source 15::1
tunnel mode gre ipv6
tunnel destination 13::1
!
router ospf 50
network 10.10.10.0 0.0.0.255 area 0
network 200.0.0.0 0.0.0.255 area 0
!
```

** Use the following commands to verify your tunnel function:

- show interface tunnel <tunnel-number>

```
Site-A-DG#show int tun 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 10.10.10.1/24
  MTU 1456 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 13::1, destination 15::1
  Tunnel protocol/transport GRE/IPV6
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Tunnel transport MTU 1456 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:02, output 00:00:07, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    43 packets input, 5472 bytes, 0 no buffer
      Received 0 broadcasts (0 IP multicasts)
        0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    48 packets output, 5899 bytes, 0 underruns
      0 output errors, 0 collisions, 0 interface resets
        0 unknown protocol drops
        0 output buffer failures, 0 output buffers swapped out
Site-A-DG#
```

- show tunnel <tunnel-number>
- show interface tunnel <tunnel-number>
statisticsTraceroute results


```

Site-A-DG#show interfaces tunnel 0 stats
Tunnel0
      Switching path      Pkts In   Chars In   Pkts Out   Chars Out
      Processor           47      5968      53      6519
      Route cache         0         0         0         0
      Total               47      5968      53      6519
Site-A-DG#

```

- Shut down Comcast-PE1 interface facing Site-A-CE1. Now Verizon becomes the active ISP between the two sites.
- Build an Automatic 6 to 4 tunnel between Site-A-CE1 and Site-B-CE2. Use IPv6 general-prefix to get the tunnel destination address automatically.

*** Hint: we don't need to specify the endpoint IPv4 address ourselves; it's determined automatically. However, these tunnels don't support IPv6 IGPs. Instead, we must use static routes.

Do a ping and traceroute from Site-A-PC3 to Site-B-PC4

Paste screenshots of the traceroute showing only two hops + commands used for building the tunnel.

Site-B-CE1:

```

interface Tunnel0
no ip address
no ip redirects
ipv6 address 2002:D00:2::2/64
tunnel source FastEthernet2/1
tunnel mode ipv6ip 6to4
!
ipv6 route 11::/64 2002:B00:2::2
ipv6 route 2002::/16 Tunnel0
ipv6 general-prefix MYPREFIX 6to4 FastEthernet2/1

```

Site-A-CE1:

```

interface Tunnel0
no ip address
no ip redirects
ipv6 address 2002:B00:2::2/64
tunnel source FastEthernet2/1
tunnel mode ipv6ip 6to4
!
ipv6 route 14::/64 2002:D00:2::2
ipv6 route 2002::/16 Tunnel0
ipv6 general-prefix MYPREFIX 6to4 FastEthernet2/1

```

Traceroute:

```

Site-A-CE1#traceroute ipv6
Target IPv6 address: 14::1
Source address: 11::1
Insert source routing header? [no]:
Numeric display? [no]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Priority [0]:
Port Number [0]:
Type escape sequence to abort.
Tracing the route to 14::1

 1 2002:D00:2::2 0 msec 0 msec 0 msec
Site-A-CE1#

```

Paste screenshot of the “show ipv6 general-prefix” result showing that the prefix is acquired automatically.

Site-A-CE1:

```

Site-A-CE1#show ipv6 general-prefix
IPv6 Prefix MYPREFIX, acquired via 6to4
 2002:B00:2::/48 Valid lifetime infinite, preferred lifetime infinite
Site-A-CE1#

```

Site-B-CE1:

```

Site-B-CE1#show ipv6 general-prefix
IPv6 Prefix MYPREFIX, acquired via 6to4
 2002:D00:2::/48 Valid lifetime infinite, preferred lifetime infinite
Site-B-CE1#

```

- Build a GRE IPv6 over IPv4 tunnel between Site-A-CE2 and Site-B-CE1. Traffic should pass through the Verizon ISP. Do a ping and traceroute from Site-A-PC4 to Site-B-PC3.
- Use Wireshark to capture the traffic passing through the tunnel.

Paste screenshot of the Wireshark capture showing the Generic Routing Encapsulation header and 0x86DD for identify IPv6 payload.

Paste Screenshot of the commands used + the status of the tunnel and traceroute results.

```

Site-B-CE1:
int lo1
ip address 3.3.3.3 255.255.255.255

router ospf 1
network 3.3.3.3 0.0.0.0 area 0

interface loopback 10
ipv6 rip GRE64 enable

```

```
!
interface tunnel 10
 tunnel source loopback 1
 tunnel destination 2.2.2.2
 tunnel mode ipv6ip
 tunnel mode gre ip
 ipv6 enable
 ipv6 rip GRE64 enable
!
ipv6 router rip GRE64

Site-A-CE2:
int lo1
ip address 2.2.2.2 255.255.255.255

router ospf 1
 network 2.2.2.2 0.0.0.0 area 0

interface loopback 10
 ipv6 rip GRE64 enable
!
interface tunnel 10
 tunnel source loopback 1
 tunnel destination 3.3.3.3
 tunnel mode ipv6ip
 tunnel mode gre ip
 ipv6 enable
 ipv6 rip GRE64 enable
!
ipv6 router rip GRE64
```

Traceroute:

```
Site-A-CE2#tracert ipv6
Target IPv6 address: 14::1
Source address: 12::1
Insert source routing header? [no]:
Numeric display? [no]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Priority [0]:
Port Number [0]:
Type escape sequence to abort.
Tracing the route to 14::1

 1 14::1 0 msec 0 msec 0 msec
Site-A-CE2#
```

*** Site-A-PC2 and Site-B-PC1 (IPv4 hosts) should not use a tunnel when Verizon is the active ISP.

*** while Comcast is down, make sure all hosts have access to the INTERNET via the Verizon ISP.

Paste screenshots of the traceroutes.

This was achieved by creating a route in automatic 6to4 that points from SiteA to SiteB so that SiteA still has internet connectivity

Site A PC4:

```
Site-A-CE2#traceroute ipv6
Target IPv6 address: 2001:db8::ffff:0808:0808
Source address: 12::1
Insert source routing header? [no]:
Numeric display? [no]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Priority [0]:
Port Number [0]:
Type escape sequence to abort.
Tracing the route to 2001:DB8::FFFF:808:808

 1 6::1 0 msec 0 msec 0 msec
 2 2002:D00:2::2 0 msec 0 msec 0 msec
 3 3::1 4 msec 0 msec 0 msec
Site-A-CE2#
```

Site B PC4:

```

Site-B-CE2#traceroute ipv6
Target IPv6 address: 2001:db8::ffff:0808:0808
Source address: 16::1
Insert source routing header? [no]:
Numeric display? [no]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Priority [0]:
Port Number [0]:
Type escape sequence to abort.
Tracing the route to 2001:DB8::FFFF:808:808

 1 9::1 0 msec 0 msec 0 msec
 2 3::1 0 msec 0 msec 0 msec
Site-B-CE2#

```

*** Turn on the interface in Comcast and make sure all hosts have access to the INTERNET via Comcast.

Paste screenshots of the traceroutes.

When I turn on Comcast, SiteA still has access to internet and now its through Comcast and not Verizon.

```

Site-A-CE2#traceroute ipv6
Target IPv6 address: 2001:db8::ffff:0808:0808
Source address: 12::1
Insert source routing header? [no]:
Numeric display? [no]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Priority [0]:
Port Number [0]:
Type escape sequence to abort.
Tracing the route to 2001:DB8::FFFF:808:808

 1 6::1 0 msec 0 msec 0 msec
 2 2::1 0 msec 0 msec 0 msec
 3 1::2 0 msec 0 msec 0 msec
Site-A-CE2#

```

*** Redistribute routes between OSPFv3 and RIPng where necessary.

*** The loopback representing internet CANNOT be advertised dynamically and it must be advertised statically (default originate).

5. Question:

Study the following tunneling mechanisms and briefly explain how each one works and its application.

1. GRE
 - a A simple generic tunnel with a source and destination IP encapsulating any packets into a GRE based packet and transferring it to the other side of the tunnel. Is used for simple one hop based communication without encryption or authentication.
2. 6 to 4 tunneling
 - a Tunnels IPv6 packets over IPv4 using the 2002::/16 prefix embedded in the IPv6. Enables isolated IPv6 networks to connect to each other over IPv4 without manual configuration.
3. 6 in 4 tunneling
 - a Encapsulates IPv6 packets in IPv4 packets using protocol 41. Requires static config
4. 6RD
 - a Like 6to4 but controlled by the ISP, using dedicated IPv6 prefix instead of 2002::/16. Lets ISPs rapidly deploy IPv6 to customers over IPv4 infra.
5. ISATAP and ISATAP prefix advertisement
 - a Emulates an IPv6 link over IPv4 within enterprise site. ISATAP hosts generate IPv6 IPs using a specific format. Enables IPv6 comms in a IPv4 corporate network. Prefix advertisements distribute said prefixes to ISATAP client routers.
6. Teredo tunneling mechanism
 - a Tunnel IPv6 over UDP/IPv4 that works through NAT devices. Uses special prefix as well. Enables IPv6 connectivity for NATed IPv4 clients.
7. mGRE
 - a Allows one GRE tunnel interface to support multiple remote endpoints. Used in DMVPN.
8. 6 in 4 Broker tunnel and Tunnel Broker operation
 - a Provides IPv6 connectivity for users with only IPv4 access. For testing or early adoption.
9. Automatic 6 to 4 tunneling
 - a Automatically creates the tunnel IP for IPv6 communication over IPv4. For easy IPv6 internet access over IPv4
10. IPv6 NAT-PT
 - a Maps an IPv4 to IPv6 like v4 NAT.
11. IPv6 Provider Edge (6PE)
 - a Carries IPv6 over MPLS backbone using IPv4 signaled LSP. IPv6 is tunneled between PE routers. Lets ISPs deliver IPv6 over IPv4/MPLS core

12. Protocol translation with NAT64/DNS64

- a NAT64 translates IPv6 client traffic to IPv4 server traffic
- b DNS64 creates quad A records from A records so IPv6 clients can reach IPv4 servers.